

Pedro SACRAMENTO X. B. ROSA

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EDUCATION

École CentraleSupélec | Master's Degree in Engineering (Bac +5)

Focused on signal processing, machine learning, robotics, and control theory.

2022 – 2025

France

University of São Paulo (Poli-USP) | Bachelor's Degree in Mechatronics Engineering

Double Degree Program.

2019 – 2025

Brazil

EXPERIENCE

Toyota Group

Research Engineer

Sep. 2025 – Present

Nice, France

- Developed signal processing pipelines for sensors (**IMU**, **LIDAR**, **Event Cameras**, etc.) using **Kalman Filter** variants to maximize information extraction from raw data.
- Designed sensor fusion techniques to integrate heterogeneous data sources, intended to feed advanced control algorithms (such as **MPC**), improving prediction robustness under real-world conditions.
- Designed low-power machine learning models (**SNNs**) in **Python** and **C++**, optimized for real-time **Edge AI** processing directly on embedded hardware.
- Configured **Linux** and **ROS2** on embedded boards, utilized **Git/GitLab** for version control, and conducted performance testing for bug resolution.
- Monitored state-of-the-art research and implemented methodologies to enhance model robustness and computational efficiency.

GE Healthcare

Software Engineer Intern - C++

Apr. 2024 – Oct. 2024

Paris, France

- Maintained and optimized medical reporting software in **C++**, ensuring high performance and reliability for over 1,000 hospital institutions.
- Integrated and benchmarked third-party **AI** tools, ensuring technical efficiency and compliance with medical regulations.
- Managed project updates and technical communications with clients (English and French) to ensure technical requirements were met.

Université de Lorraine, LMOPS

Research Intern

Jun. 2023 – Aug. 2023

Metz, France

- Developed a modular **Python** and **PyTorch** framework for control tasks, implementing **Reinforcement Learning** (A2C, REINFORCE) and evaluating **Model Predictive Control (MPC)** approaches.
- Optimized computational architecture by replacing traditional networks with Echo State Networks (**ESNs**) to efficiently model nonlinear dynamical systems.
- Benchmarked models on Atari and classic-control environments, achieving a **66% reduction** in training time and a **15% increase** in sampling efficiency.

SKILLS

Languages : Portuguese (Native), English (Fluent), French (Fluent)

Programming : C++, Python, C, Matlab

Algorithms & Theory : Sensor fusion, state estimation (Kalman Filters), signal processing, control theory, model predictive control (MPC), reinforcement learning (RL), real-time Edge AI

Technologies : ROS2, Linux, Embedded Systems, TensorFlow, PyTorch, Spiking Neural Networks (SNNs), Git, CI/CD, Docker, Kubernetes